Problem 3 - Orchestra Warm-Up Routine

Week 7 | Category B Practice | Java solutions

Problem

Instrument <- StringInstrument <- Violin. Concrete overrides build their messages by calling super.play() first.

Complete Java Code

```
abstract class Instrument {
    public abstract String play();
}

class StringInstrument extends Instrument {
    public StringInstrument() { super(); }
    @Override
    public String play() {
        return "Strumming the strings";
    }
}

class Violin extends StringInstrument {
    public Violin() { super(); }
    @Override
    public String play() {
        return super.play() + ", with a bow drawn across four strings";
    }
}

public class Main {
    public static void main(String[] args) {
        StringInstrument s = new StringInstrument();
        Violin v = new Violin();
        System.out.println(s.play());
        System.out.println(v.play());
    }
}
```

Sample Output

```
Strumming the strings
Strumming the strings, with a bow drawn across four strings
```

How it works

The parent abstract class requires play(). StringInstrument supplies the general message, and Violin extends it by calling super.play() and adding its bow detail.

Source basis: Week 7 Category B assignment/practice PDF. Examples and requirements are preserved; implementation details are filled in to create a complete runnable Java example.